11 When a cork is pushed down into still water in a tank and then released, it oscillates with simple harmonic motion of period 0.12 s.

Water waves are then set up in the tank and its frequency adjusted until the cork is observed to have a maximum amplitude of vibration.

The speed v of the water waves is given by

$$v = \sqrt{gd}$$

where g is the acceleration due to gravity and d is the depth of the water in the tank.

If the depth of the water in the tank is 0.30 m, what is the wavelength of the water waves when the cork is moving up and down with the largest amplitude?

A 0.070 m **B** 0.14 m **C** 0.21 m **D** 0.35 m